NovaCorp ValueGuard

Core story

NovaCorp knows who it cannot afford to lose. It does not yet know the actionable reason or correct intervention before those employees leave. ValueGuard closes that decision gap through a controlled, voluntary Stay-and-Act pilot and a separately governed Agency Value Gate.

Decision	Base planning case	Critical boundary
Approve a capped controlled experiment	$0.500M envelope; $2.133M illustrative net	Concentration is verified; causation and savings are not

Single narrative spine

Beat	Question answered	Answer
Reveal	Where is the value at risk?	9.03% of employees carry 49.62% of historical regrettable replacement exposure.
Challenge	Do we know why?	No universal pre-exit mechanism survived timing, competing-explanation and stability gates.
Resolve	What should management do?	Ask before prescribing; match employee-stated friction to bounded action.
Prove	How do we learn whether it works?	Controlled offer assignment, ITT, comparison, 6/12-month outcomes and Finance validation.
Decide	What is requested now?	Approve capped pilot + Agency Value Gate; scale, adapt or stop based on evidence.

Financial case

Three equivalent avoided HiPo losses roughly cover the base pilot.

- Average HiPo regrettable replacement exposure = $12.634M ÷ 76 = ~$166,240 per event.
- Base pilot break-even = $500,000 ÷ ~$166,240 = 3.01 equivalent incremental HiPo regrettable losses avoided.
- Retention gross = $12.634M × 10% = $1.263M.
- Agency gross = $4.565M × 30% = $1.370M.
- Total gross = $2.633M; less $0.500M programme envelope = $2.133M net.
- ROI = $2.133M ÷ $0.500M = 427%; payback = $0.500M ÷ $2.633M × 12 = 2.28 months.

Close and implementation

- 0–30 days: capture HiPo designation history and regret criteria; freeze eligibility/assignment; approve privacy, strata, action menu, cost cap, stop rules and agency exceptions.
- 31–90 days: assign voluntary offers; conduct conversations; complete bounded actions; audit protocol, trust, fairness and cost.
- 3–6 months: adapt or stop if integrity, trust or execution fails; lock the outcome analysis.
- 6–12 months: estimate voluntary-retention difference, uncertainty and verified net value; scale, adapt or stop.
- Owners: CHRO sponsor; Head of Talent/Analytics; Privacy + employee representative; business action owners; CFO; Talent Acquisition.
- Final ask: APPROVE VALUEGUARD as a controlled experiment—not a permanent programme.

Rules and guardrails applied for data cleaning and testing:

Category	Guardrail or rule	How it was applied	What it prevents
Time	Outcome cutoff rule	For leavers, the cutoff is their exit date. For active employees, it is 31 December 2025.	Information from after the decision point influencing historical analysis.
Data leakage	Predictor availability rule	A variable can be used prospectively only if it was known before the employee's cutoff date.	Target leakage and unrealistically strong models.
Data leakage	Post-exit fields blocked	regrettable_flag, exit reason, pathway and performance band at exit were outcomes or descriptive labels only.	Using future HR classifications to predict the event that created them.
Data leakage	Exit-process fields blocked	Notice served, manager at exit and salary at exit were not used for pre-resignation targeting. Salary at exit was used only for cost estimation.	Exit-process information being mistaken for an early-warning signal.
Time	Survey-before-outcome rule	Only survey opportunities dated before the outcome cutoff were used. Six-month outcomes required adequate follow-up.	Post-exit surveys, immature observations and reverse sequencing.
Time	Review-before-outcome rule	Performance and promotion information was used only when the review date	Later performance information explaining an

Category	Guardrail or rule	How it was applied	What it prevents
		preceded the relevant survey or exit.	earlier event.
Time	Promotion pathway sequencing	The tested sequence was: dated review → next eligible engagement wave → later exit.	Claiming a career mechanism without correct event order.
Time	HiPo timing limitation	HiPo designation dates were unavailable, so HiPo was used as a formal segment—not a historically time-varying causal feature.	Claiming that HiPo designation preceded or caused exit.
Time	Snapshot-variable restriction	Department, manager, role, level, acting appointment, promotion eligibility and compensation were treated as current snapshots.	Assuming current organisational conditions existed before every historical exit.
Time	Tenure reconstruction	tenure_months had to be reconstructed at the relevant index date rather than treated as a fixed baseline value.	Outcome-dependent tenure leakage.
Follow-up	Mature outcome rule	Hiring retention analyses required complete 3-, 6- or 12-month follow-up.	Counting recent hires as retained merely because they had insufficient observation time.
Definitions	HiPo definition guardrail	hipo_flag means formal talent-review designation. It is not synonymous with high performance.	Misrepresenting the official business definition.
Definitions	Regrettable-exit definition	regrettable_flag is an HR classification made after voluntary departure.	Treating a subjective retrospective classification as objective prospective truth.
Definitions	Promotion recommendation rule	A recommendation is not an actual promotion; false does not mean promotion was denied.	Inventing employee career events that are absent from the data.
Definitions	Survey silence rule	response_flag = 0 means no response to an eligible survey opportunity—not disengagement or resignation intent.	Labelling employees as disengaged or likely to resign.
Definitions	Goal-score limitation	Goal achievement was treated as a performance proxy, not direct operational productivity.	Converting an HR score directly into productivity dollars.
Definitions	Acquisition-origin limitation	legacy_entity_code identifies acquisition origin/context, not a causal reason for attrition.	Blaming acquired entities or employees for observed differences.
Population	Denominator discipline	Workforce concentration used 13,403	Mixing denominators to

Category	Guardrail or rule	How it was applied	What it prevents
		employees; regrettable-event concentration used 153 exits; agency analysis used 274 agency hires.	create misleading percentages.
Population	Survey eligibility enforcement	Only employees genuinely eligible for a particular survey wave entered that wave's denominator.	Artificially depressing response rates through ineligible employees.
Comparison design	Exit-aligned comparison	Historical HiPo cases were aligned to comparable points in time and compared with active HiPo stayers.	Comparing future information for leavers with current information for stayers.
Comparison design	Matching guardrail	The 76 HiPo regrettable leavers were matched to 304 active HiPo stayers; standardised differences checked balance.	Obvious observed composition differences driving the comparison.
Causal inference	Association ≠ causation	Even adjusted results were described as associations unless timing, identification and confounding requirements were satisfied.	Turning statistical significance into unsupported causal claims.
Causal inference	Missing-variable boundary	Matching and regression were explicitly recognised as unable to correct missing designation dates, external offers or unmeasured employee reasons.	Presenting adjusted models as causal proof.
Causal inference	No universal-root-cause claim	Pay, promotion, engagement, manager, role and entity mechanisms were tested but none was declared the universal cause.	Forcing several weak signals into one convenient causal story.
Specificity	Negative-control test	Survey non-response was tested against involuntary exit as well as voluntary exit. Both showed associations.	Claiming silence specifically signals resignation when it reflects broader instability.
Operational usefulness	Precision/PPV rule	Predictive performance was assessed using precision and PPV—not ROC-AUC alone. Regrettable-exit PPV was only 0.39%.	Deploying a model that ranks acceptably but falsely flags almost everyone.
Model purpose	Prediction ≠ intervention	A risk-ranking model was not treated as evidence that a particular intervention will work.	Confusing prediction with treatment effectiveness.
Model validity	Events-per-parameter check	Sparse adjusted models were treated as sensitivity analyses where events per parameter were limited.	Overconfident interpretation of unstable regression estimates.
Model validity	Convergence and conditioning	Convergence, event sparsity and condition-number diagnostics were used	Reporting numerically fitted but unreliable models.

Category	Guardrail or rule	How it was applied	What it prevents
		to identify unstable specifications.	
Repeated data	Cluster-robust uncertainty	Repeated reviews or survey observations from the same employee used clustered or robust standard errors.	Treating repeated records as independent people.
Robustness	Non-parametric testing	Mann–Whitney U, Cliff's delta, median regression or bootstrap methods were used where normality assumptions were unsuitable.	Relying on inappropriate mean-based tests for skewed data.
Robustness	Survival diagnostics	Kaplan–Meier, log-rank and Cox sensitivity analyses were accompanied by proportional-hazards checks.	Reporting hazard ratios from a violated model without qualification.
Multiple testing	Multiplicity correction	Holm or Benjamini–Hochberg corrections were used where many related hypotheses were tested.	Treating chance discoveries as confirmed findings.
Sparse cells	Fisher exact rule	Fisher's exact test was preferred for sparse 2×2 comparisons.	Invalid chi-square approximations in small samples.
Sparse cells	Zero-cell correction	A 0.5 continuity correction was used for zero cells where an odds ratio would otherwise become infinite.	Dramatic but meaningless infinite estimates.
Sparse cells	Small-cell suppression	Tiny intersections were suppressed, downgraded or excluded from headline targeting.	Re-identification, unstable targeting and exaggerated subgroup claims.
Effect reporting	Magnitude with uncertainty	Risk difference, risk ratio, odds ratio, confidence intervals and effect sizes accompanied p-values.	"Statistically significant" being presented without practical magnitude.
Entity interpretation	Confounding by integration history	Entity, tenure, HRIS transition and acquisition timing were recognised as entangled.	Presenting Entity_B or Entity_C membership as an independent cause.
Fairness	Protected attributes excluded from targeting	Gender, age band and cultural background cannot determine pilot selection, offers or actions.	Discriminatory employee targeting.
Fairness	Aggregate fairness audit only	Protected attributes may be used only to audit representation, offers, actions and outcomes at aggregate level.	Using fairness data as a selection mechanism.
Fairness	Multi-stage disparity monitoring	Fairness is checked at selection, offer, uptake, action allocation, completion and outcome stages.	A fair-looking initial sample hiding unequal delivery or outcomes.

Category	**Guardrail or rule**	**How it was applied**	**What it prevents**
Fairness	Material unexplained disparity stop rule	The pilot pauses if a material disparity cannot be explained and corrected.	Scaling an intervention that produces unequal treatment or harm.
Privacy	Data minimisation	Only the minimum fields needed for assignment, delivery, evaluation and cost validation are collected.	Building an unnecessary employee-surveillance dataset.
Privacy	Separate records	Pilot records should remain separate from performance-management records.	Retention conversations affecting appraisal or disciplinary processes.
Privacy	Role-based access	Access is restricted according to job responsibility. Managers do not receive individual survey answers.	Unauthorised viewing or misuse of sensitive employee information.
Privacy	Audit logs	Access and material decisions are logged.	Undetectable browsing, alteration or misuse of employee data.
Privacy	Retention/deletion schedule	Pilot data must have an approved deletion schedule.	Sensitive pilot information being retained indefinitely.
Privacy	Purpose limitation	Information collected for ValueGuard cannot quietly expand into performance, promotion or disciplinary uses.	Function creep.
Employee autonomy	Voluntary participation	Stay conversations and employee participation are voluntary.	Coercive retention practices.
Employee autonomy	Clear notice	Employees must be told the purpose, data use, confidentiality boundaries and voluntary nature of the pilot.	Hidden monitoring or deceptive data collection.
Employee autonomy	Employee-stated friction only	Actions are selected from concerns the employee voluntarily identifies.	Inferring health, family, union status, personal circumstances or resignation intent.
Employee autonomy	Correction and appeal	Employees can correct information and challenge decisions or records.	Acting on inaccurate or misunderstood employee data.
Employee autonomy	Normal support outside pilot	Employees who decline or are not assigned must retain access to ordinary HR and management support.	Penalising non-participation.
Human oversight	No automated adverse decisions	ValueGuard cannot autonomously make pay, promotion, termination, performance or disciplinary decisions.	Algorithmic harm from uncertain signals.

Category	Guardrail or rule	How it was applied	What it prevents
Human oversight	No flight-risk ranking	The proposed solution uses a controlled offer assignment—not an individual resignation-risk score.	Creating employee watchlists.
Human oversight	No manager punishment	Manager-level signals cannot automatically trigger blame or sanctions.	Punishing managers based on sparse or confounded statistics.
Pilot design	Controlled offer assignment	Eligibility, strata, exclusions and controlled/random assignment must be frozen before delivery.	Managers selecting favourite employees or highest-looking “risk” cases.
Pilot design	Balanced strata	Material tenure, role-level and department strata inform balanced pilot coverage, not causal targeting.	Overconcentrating the pilot in one visible group and confusing strata with causes.
Pilot evaluation	Intention-to-treat analysis	Outcomes are analysed according to original offer assignment, even when employees decline or actions are incomplete.	Selective participation making the intervention appear more effective.
Pilot evaluation	Comparison group	Retention outcomes require an assigned comparison, not just before-versus-after observation.	Attributing normal changes or external trends to ValueGuard.
Pilot evaluation	Pre-specified analysis	Eligibility, assignment, outcomes, exclusions and analysis rules are frozen before results are known.	Moving thresholds or rewriting success criteria after seeing results.
Pilot evaluation	Outcome hierarchy	Voluntary retention is the main prospective outcome; regret is secondary because it is rare and classified after exit.	Building the evaluation around a sparse, retrospective label.
Pilot evaluation	Process before impact	Day 90 is a delivery, trust, fairness and protocol gate—not a savings or retention-effect claim.	Claiming success before enough outcome time exists.
Pilot evaluation	6/12-month outcome window	Incremental retention is assessed at six and twelve months.	Reporting premature retention effects.
Stop rules	Retaliation	Stop for substantiated retaliation linked to participation or disclosed concerns.	Employee harm.
Stop rules	Coercion	Stop if participation or acceptance of actions becomes coercive.	Forced engagement presented as programme success.
Stop rules	Unauthorised disclosure	Stop for material confidentiality or access breaches.	Continuing after a loss of employee trust or privacy.
Stop rules	Purpose expansion	Stop if data or processes expand beyond	Surveillance and function

Category	Guardrail or rule	How it was applied	What it prevents
		their approved purpose.	creep.
Stop rules	Budget breach	Pause when actions exceed the approved programme envelope or delegated authority.	Uncontrolled spending.
Stop rules	Service degradation	Stop or adapt if the intervention damages operations or employee service.	Retention activity creating broader organisational harm.
Financial	Historical proxy language	Replacement cost is called a historical gross exposure or replacement-cost proxy—not audited loss.	Presenting modelled exposure as realised cash expenditure.
Financial	Scenario—not-forecast rule	Conservative, base and upside results must be labelled planning scenarios.	Claiming ValueGuard will definitely save $2.133M.
Financial	No observed treatment effect	The repo states that ValueGuard effectiveness has not yet been observed.	Turning assumed capture percentages into evidence of success.
Financial	Transparent retention formula	HiPo exposure = salary at exit × 1.5 replacement multiplier × 85% attribution. Scenario benefit = $12.634M × 5%/10%/15%.	Hiding assumptions behind a headline savings number.
Financial	Transparent agency formula	Agency fee scenario = salary base × 18%; direct comparator = hires × $5,500; premium = difference.	Presenting the $4.565M premium as audited invoices.
Financial	Addressable-volume rule	Only substitutable/default agency volume is considered addressable.	Assuming every agency placement can or should be removed.
Financial	No agency-ban claim	The proposal governs and measures agency exceptions; it does not claim agencies are always inefficient.	Ignoring scarce-role complexity and legitimate specialist hiring.
Financial	Disengagement excluded	No disengagement-productivity benefit is included in ValueGuard scenarios.	Monetising an unresolved definition without direct productivity measurement.
Financial	No early-exit double counting	Hiring early-exit replacement benefits are not added to the agency case.	Counting the same economic loss twice.
Financial	Pools not blindly summed	Attrition, disengagement and hiring estimates use different definitions and may overlap.	Claiming the entire $42M is independently measurable and recoverable.
Financial	Net-benefit formula	Net = retention gross + agency gross – programme cost. ROI = net ÷ programme	Reporting gross exposure as net financial value.

Category	Guardrail or rule	How it was applied	What it prevents
		cost.	
Financial	Finance recognition rule	Value is recognised only for demonstrated avoided invoices/backfills, actual programme costs and validated non-overlap.	Treating scenario arithmetic as realised savings.
Financial	Payback qualification	Payback is straight-line scenario arithmetic; actual timing depends on when costs are genuinely avoided.	Promising a guaranteed payback date.
Claims	Evidence-status labels	Claims are classified as verified, supported association, unresolved, not validated, blocked by data or scenario—not forecast.	Presenting every finding with the same certainty.
Claims	No causal stitching	HiPo concentration, survey silence, pay, career and manager findings cannot be combined into one unsupported causal pathway.	Creating a persuasive but scientifically unsupported story.
Governance	Defined accountability	CHRO sponsors; Talent operates; executives deliver actions; Analytics evaluates; CFO validates value; Privacy and employee representatives govern.	Unclear ownership and uncontrolled decision-making.
Governance	Scale/adapt/stop gate	Scaling requires positive incremental value, implementation fidelity, acceptable trust and no material fairness harm.	Expanding based only on participation or attractive scenario numbers.
Governance	Prospective-data requirement	HiPo designation history, actual employee reasons, transfer history, external offers, invoices and direct productivity require new data.	Pretending further retrospective modelling can answer unobserved questions.

Core defence appendix .

13 — What survived every evidence gate—and what did not

Claim	Status	What may be said	Decision consequence
HiPo cost concentration	VERIFIED	9.03% workforce; 49.62% historical exposure	Use formal HiPo cohort for governed pilot.
Formal HiPo association	SUPPORTED ASSOCIATION	6.28% vs 0.63% regrettable-exit rate; timing absent	Do not call HiPo status causal.
Survey nonresponse	SUPPORTED, NONSPECIFIC	OR 2.25 voluntary; OR 2.12 involuntary	Listening/access only; no flight-risk score.
Pay/career/manager common pathway	NOT VALIDATED / BLOCKED	May matter individually; no universal treatment	Employee-stated action only.
Disengagement dollars	NOT VALIDATED	Definition-sensitive Finance scenarios	Exclude from financial case.
ValueGuard savings	UNKNOWN	Prospective treatment effect required	Controlled experiment and ITT.

Sources: https://github.com/MoinIhtejaz/Accenture_Datathon/blob/Raiha/docs/final_strategy/evidence_register.md | https://github.com/MoinIhtejaz/Accenture_Datathon/blob/Raiha/docs/final_storyline.md

14 — Every pain point has a decision answer

Pain point	Observed	Evidence verdict	Action
Entity_B	~62–65% eligible response	Pressure pocket; cause confounded	Repair access/listening; represent in pilot.
Entity_C	68.6% in one eligible wave	Insufficient history	Monitor prospectively.
R&C / FAR	$3.811M / 20 HiPo events	Material; mechanism unresolved	Balance coverage; collect workload/market/FAR evidence.
Career	Broad plausibility	No universal dated pathway	Use only when employee-stated.
Pay / manager	Snapshot or weak timing	Causal targeting blocked	Formal review/support only when relevant.
Silence	Time-ordered association	Signal, not intent	Operational listening check only.

Sources: https://github.com/MoinIhtejaz/Accenture_Datathon/blob/Raiha/docs/final_report/tables/table_03_hypothesis_verdicts.csv

15 — Known, unknown and what ValueGuard learns

Known now	Not known now	ValueGuard is designed to learn
Where historical exposure is concentrated	Why a particular employee may leave	Employee-stated friction before exit
No universal mechanism validated	Which action changes retention	Reason–action combinations and completion
Current predictive precision is unsafe	True causal treatment effect	ITT retention difference at 6/12 months
Agency premium scenario is material	Actual avoidable invoices/backfills	Finance-verified incremental net value
Required ethical boundaries	Whether trust/fairness remains acceptable	Trust, fairness, misuse and stop-rule evidence

Sources: https://github.com/MoinIhtejaz/Accenture_Datathon/blob/Raiha/docs/final_storyline.md | https://github.com/MoinIhtejaz/Accenture_Datathon/blob/Raiha/docs/final_strategy/implementation_plan.md

Financial claim construction and formula audit

Non-negotiable language

All values below are planning proxies or decision scenarios. None is an audited cash loss, forecast, guaranteed saving or observed treatment effect. Finance recognition requires actual avoided invoices/backfills, actual programme cost and removal of overlap.

A. Historical regrettable replacement exposure

Event-level formula: salary at exit × 1.50 replacement multiplier × 0.85 backfill/attribution factor.

Population	Events	Calculated exposure	Share / use
All HR-regrettable voluntary exits	153	$25,462,005	Gross extract-derived proxy; $0.462M above official $25M range.
Recorded formal HiPos	76	$12,634,230	49.62% of all exposure.
Not recorded HiPo	77	$12,827,775	50.38% of all exposure.

Average HiPo event exposure = $12,634,230 ÷ 76 = $166,239.87.

Cost concentration multiple = 49.6199% exposure share ÷ 9.0278% workforce share = 5.4963×.

Sources: https://github.com/MoinIhtejaz/Accenture_Datathon/blob/Raiha/outputs/tables/hipo_root_cause/hipo_outcome_concentration.csv | https://github.com/MoinIhtejaz/Accenture_Datathon/blob/Raiha/docs/final_report/tables/table_01_hipo_concentration.csv

B. Agency premium scenario

Step	Formula	Result
Agency fee scenario	$33,734,300 observed salary base × 18%	$6,072,174
Direct benchmark	274 observed agency hires × $5,500	$1,507,000
Implied premium	$6,072,174 − $1,507,000	$4,565,174

Boundary: the repo does not contain actual agency invoices, requisition complexity, scarcity value or a causal quality comparison. The gate should govern exceptions, not ban agencies.

Sources: https://github.com/MoinIhtejaz/Accenture_Datathon/blob/Raiha/docs/final_report/tables/table_04_agency_economics.csv | https://github.com/MoinIhtejaz/Accenture_Datathon/blob/Raiha/outputs/tables/hiring_economics_retest.csv

C. Scenario formulas

Retention gross = $12,634,230 × assumed capture. Agency gross = $4,565,174 × assumed premium reduction. Net benefit = retention gross + agency gross − programme envelope. ROI = net benefit ÷ programme envelope. Payback months = programme envelope ÷ total gross × 12.

Scenario	Retention assumption	Agency assumption	Cost	Gross	Net	ROI	Payback
Conservative	5% = $631,712	0% = $0	$250,000	$631,712	$381,712	153%	4.75 mo
Base	10% = $1,263,423	30% = $1,369,552	$500,000	$2,632,975	$2,132,975	427%	2.28 mo
Upside	15% = $1,895,135	50% = $2,282,587	$750,000	$4,177,722	$3,427,722	457%	2.15 mo

Sources: https://github.com/MoinIhtejaz/Accenture_Datathon/blob/Raiha/docs/final_report/tables/table_05_financial_scenarios.csv

D. What must not be added

- No disengagement productivity value: six defensible definitions generate $10.606M, $45.379M, $106.046M, $32.583M, $79.832M and $11.016M under the same 15% Finance assumption.
- No agency early-exit replacement value unless it is linked, incremental and de-duplicated from the attrition pool.
- No value for all retained employees: count only an incremental comparison-based effect attributable to original offer assignment.
- No promised payback date: the current payback figure is scenario arithmetic, not realised timing.

Sources: https://github.com/MoinIhtejaz/Accenture_Datathon/blob/Raiha/outputs/tables/value_leakage_analysis.csv

Evidence Lab: statistical defence appendix (A1–A19)

Use the same seven-part rail on every appendix slide: Business question → Design → Method → Result → Robustness → Decision boundary → Ethics link.

A1 — Evidence audit before testing

BUSINESS QUESTION

Can the extracts support the question?

DESIGN / METHOD

Controlled cleaning, source manifests, population definitions and validation checks.

REPO-GROUNDED RESULT

13,403 employees; dated and snapshot fields separated; defects and missing timing logged.

DECISION BOUNDARY / ETHICS LINK

Data quality is a statistical assumption; invalid definitions cannot become business findings.

A2 — Definition and timing gate

BUSINESS QUESTION

What does each variable actually mean and when is it known?

DESIGN / METHOD

Feature-definition audit and temporal leakage audit.

REPO-GROUNDED RESULT

HiPo is formal snapshot designation; regret is post-exit; response is eligible-opportunity behaviour; recommendation is not promotion.

DECISION BOUNDARY / ETHICS LINK

Blocks causal or predictive uses that the data cannot support.

A3 — Beyond p-values

BUSINESS QUESTION

Did the finding have magnitude, uncertainty and stability?

DESIGN / METHOD

Effect sizes, confidence intervals, adjusted models, dependence correction, multiplicity, falsification and sensitivity.

REPO-GROUNDED RESULT

Only results clearing pre-specified evidence gates influenced the decision.

DECISION BOUNDARY / ETHICS LINK

A small p-value alone never determines strategy.

A4 — HiPo concentration

BUSINESS QUESTION

Is the HiPo difference large in absolute and relative terms?

DESIGN / METHOD

2×2 rates, risk difference, RR/OR, logistic sensitivity and bootstrap cost-share interval.

REPO-GROUNDED RESULT

6.28% vs 0.63%; +5.65pp; OR 10.55 (95% CI 7.64–14.56); cost share 49.62% (bootstrap 41.42–57.86%).

DECISION BOUNDARY / ETHICS LINK

Large association; designation timing blocks causation.

A5 — Matched HiPo comparison

BUSINESS QUESTION

Can comparable historical cases and stayers be evaluated at the same time?

DESIGN / METHOD

76 cases matched to 304 active stayers; standardised differences assess balance.

REPO-GROUNDED RESULT

76 matched case sets; no reliable universal divergence feature.

DECISION BOUNDARY / ETHICS LINK

Matching reduces observed imbalance but cannot fix missing dates/unmeasured confounding.

A6 — Temporal leakage control

BUSINESS QUESTION

Did future information enter the past?

DESIGN / METHOD

Exit-aligned eligibility windows and explicit post-cutoff blocks.

REPO-GROUNDED RESULT

0 post-cutoff events used; 0 invented HiPo dates; validation 11/11 passed.

DECISION BOUNDARY / ETHICS LINK

Protects historical inference from target leakage.

A7 — Promotion pathway

BUSINESS QUESTION

Does recommendation/promotion timing create a common pathway?

DESIGN / METHOD

Dated review → next eligible engagement wave → later exit; OLS/robust covariance and time-aligned models.

REPO-GROUNDED RESULT

Engagement composite change +0.006; pathway mixed or null.

DECISION BOUNDARY / ETHICS LINK

Career support remains available only when employee-stated.

A8 — Survey silence

BUSINESS QUESTION

Is nonresponse a useful resignation signal?

DESIGN / METHOD

Employee-wave cluster-robust logistic model; marginal RD; involuntary negative control; PPV.

REPO-GROUNDED RESULT

Voluntary OR 2.25; adjusted RD ~+2.30pp; involuntary OR 2.12; PPV 4.40% voluntary and 0.39% regrettable.

DECISION BOUNDARY / ETHICS LINK

Listening/access check only; no individual flight-risk score.

A9 — Sparse cells

BUSINESS QUESTION

How were small or zero cells handled?

DESIGN / METHOD

χ^2 + Cramér's V for larger tables; Fisher exact for sparse 2×2; 0.5 correction for zero cells; suppression/downgrade.

REPO-GROUNDED RESULT

L4: 21 HiPos / 2 events; small manager teams underpowered.

DECISION BOUNDARY / ETHICS LINK

Uncertainty becomes a privacy and no-targeting rule.

A10 — R&C sensitivity

BUSINESS QUESTION

Does R&C support a specific intervention?

DESIGN / METHOD

Adjusted interactions plus survival sensitivity and assumption diagnostics.

REPO-GROUNDED RESULT

R&C × senior OR 3.26 (1.41–7.53), low-cell; R&C × below-market compa OR 1.12 (0.77–1.62), unsupported.

DECISION BOUNDARY / ETHICS LINK

Include for balanced coverage and collect FAR/workload/market evidence; no automatic pay action.

A11 — Hiring economics

BUSINESS QUESTION

Are agencies faster or better enough to justify default premium use?

DESIGN / METHOD

Mann–Whitney U, Cliff's delta, adjusted median regression and 1,000 employee bootstrap samples.

REPO-GROUNDED RESULT

Median days: agency 53, direct 56, referral 56.5; superiority unresolved.

DECISION BOUNDARY / ETHICS LINK

Gate and document exceptions rather than ban agencies.

A12 — Prediction not deployment

BUSINESS QUESTION

Does model performance justify individual action?

DESIGN / METHOD

AUC, Brier, 5-fold stratified CV, top-decile precision, events/parameter, condition number and convergence.

REPO-GROUNDED RESULT

Diagnostic ranking does not establish usable precision, treatment value or prospective validity.

DECISION BOUNDARY / ETHICS LINK

No automated individual retention score.

A13 — Multiplicity

BUSINESS QUESTION

Could one lucky p-value drive strategy?

DESIGN / METHOD

Holm for registered/confirmatory families; Benjamini–Hochberg for exploratory signals; effect-size and stability gates.

REPO-GROUNDED RESULT

No pre-exit feature met all reliable-divergence gates.

DECISION BOUNDARY / ETHICS LINK

Prevents selective finding promotion.

A14 — Sensitivity analysis

BUSINESS QUESTION

Does a conclusion survive reasonable alternative assumptions?

DESIGN / METHOD

Wave, timing, age, cohort, model, matching, definition and capture-rate sensitivity.

REPO-GROUNDED RESULT

Silence survives association checks but fails specificity; R&C mechanism remains unstable; disengagement dollars change drastically by definition.

DECISION BOUNDARY / ETHICS LINK

Only stable conclusions enter the recommendation.

A15 — Disengagement definition

BUSINESS QUESTION

Can the $12–15M productivity pool be reproduced uniquely?

DESIGN / METHOD

Six candidate definitions with the same 15% Finance formula; composite reliability test.

REPO-GROUNDED RESULT

$10.606M to $106.046M; Cronbach α 0.749 across 45,707 complete respondent-waves.

DECISION BOUNDARY / ETHICS LINK

A coherent score does not validate a business threshold or direct productivity loss.

A16 — Model self-correction

BUSINESS QUESTION

What happened when a model specification failed audit?

DESIGN / METHOD

Symmetric trajectory rebuild and specification sensitivity.

REPO-GROUNDED RESULT

Defective/asymmetric narrative removed; corrected pattern retained only as a limited burden proxy.

DECISION BOUNDARY / ETHICS LINK

Self-correction is evidence governance, not weakness.

A17 — Evidence-derived ethics

BUSINESS QUESTION

How did statistics create the guardrails?

DESIGN / METHOD

Map association, negative control, PPV and data limitations to operating rules.

REPO-GROUNDED RESULT

Broad signal + low specificity → listening only; snapshot pay → no pay targeting; post-exit regret → no worth score.

DECISION BOUNDARY / ETHICS LINK

Ethics rules are mathematically and operationally motivated.

A18 — Fairness separate from targeting

BUSINESS QUESTION

How are protected characteristics used?

DESIGN / METHOD

Aggregate representation, offer, action and outcome disparity audits with suppression.

REPO-GROUNDED RESULT

Gender, age band and cultural background are excluded from intervention selection.

DECISION BOUNDARY / ETHICS LINK

Fairness audit only; never targeting.

A19 — Prospective treatment effect

BUSINESS QUESTION

How will ValueGuard learn whether it works?

DESIGN / METHOD

Controlled offer assignment, original-assignment ITT, comparison, 6/12-month retention and Finance verification.

REPO-GROUNDED RESULT

Historical analysis identifies the decision gap; the pilot estimates incremental effect.

DECISION BOUNDARY / ETHICS LINK

ITT limits self-selection bias and discourages coercion; scale/adapt/stop gates remain mandatory.

Statistical methods inventory: what was done and why

Method	Where used	Purpose
Risk difference + CI	Absolute outcome difference	Translate association to percentage points.
Risk ratio / odds ratio + CI	Relative binary association	Magnitude and uncertainty, not p-value alone.
χ^2 + Cramér's V	Categorical dependence	Test association and practical effect size.
Fisher exact + 0.5 correction	Sparse/zero 2×2 cells	Avoid invalid asymptotics or infinite ORs.
Logistic regression	Adjusted binary outcomes	Control observed covariates; produce adjusted associations.
Marginal adjusted RD	Model-to-business translation	Return adjusted model to percentage-point impact.
Cluster-robust SE	Repeated employee records	Account for within-employee dependence.
OLS + HC0/HC3	Continuous change	Estimate continuous pathway with robust covariance.
Mann–Whitney U + Cliff's δ	Non-normal/ordinal comparisons	Distribution-robust test plus effect size.
Median regression + bootstrap	Hiring fill time	Adjusted median comparison and sampling uncertainty.
Kaplan–Meier / log-rank	Time to exit	Compare event timing, not only binary occurrence.
Cox + PH diagnostics	Adjusted survival sensitivity	Estimate hazards and test proportional-hazard assumption.
Holm / Benjamini–Hochberg	Multiple hypotheses	Control family-wise error / false discoveries.
Standardised differences	Matching balance	Check case/control comparability.
Negative controls	Falsification/specificity	Test whether a supposed resignation signal is broader.
Cronbach's α	Composite reliability	Check internal consistency, not threshold validity.
Cross-validated ridge logistic	Predictive diagnostic	Separate ranking performance from treatment usefulness.
AUC / Brier / precision	Model diagnostics	Discrimination, probability error and operational usability.
Condition number / EPP / convergence	Model validity	Detect instability, sparsity and failed estimation.
Sensitivity analysis	Robustness	Ask whether decisions change under reasonable assumptions.

Sources: https://github.com/MoinIhtejaz/Accenture_Datathon/blob/Raiha/scripts/phase_f_statistical_validation.py | https://github.com/MoinIhtejaz/Accenture_Datathon/blob/Raiha/docs/final_report/appendix/technical_evidence_appendix.md

Implementation, governance and measurement handover

90-day build and 12-month evidence horizon

Timing	Actions	Evidence gate
Days 0–30	Designation history; regret criteria; eligibility; assignment; privacy; strata; action menu; cost cap; stop rules; agency exceptions.	Design is frozen, lawful, auditable and finance-owned.
Days 31–60	Controlled offers; operational checks; voluntary conversations; friction/action/owner/cost/consent records.	Protocol integrity, uptake, confidentiality and action feasibility.
Days 61–90	Completion/follow-up; trust, fairness, cost and misuse audit.	Continue only if delivery and employee safeguards pass.
Months 3–6	Adapt/stop review; lock 6/12-month outcome specification.	No metric switching or post-hoc subgroup hunting.
Months 6–12	ITT retention difference, uncertainty, verified avoided cost and actual programme cost.	Scale, adapt or stop on incremental net value and no material harm.

Owner and KPI map

Executive question	KPI	Timing	Owner
Did we run the design correctly?	Eligibility, assignment and offer-protocol integrity	Weekly	Head of Talent / Analytics
Do employees trust it?	Uptake, usefulness, confidentiality/coercion complaints	30/60/90	Privacy + employee representative
Did actions happen?	Completion on time and within approved limit	Monthly	Business executive
Did retention improve?	6/12-month voluntary retention vs comparison	6/12 months	People Analytics
Was value protected?	Verified avoided cost less actual programme cost	Quarterly	CFO
Is agency spend disciplined?	Exceptions, premium, fill time, quality, retention	Monthly/quarterly	Talent Acquisition + CFO
Was it fair?	Selection, offer, action and outcome disparities	Monthly/quarterly	Privacy + employee representative

Sources: https://github.com/MoinIhtejaz/Accenture_Datathon/blob/Raiha/docs/final_report/tables/table_07_kpis.csv | https://github.com/MoinIhtejaz/Accenture_Datathon/blob/Raiha/docs/final_strategy/implementation_plan.md

Repository source map

Use	Repository path / URL	Use	Repository path / URL
Final executive storyline	https://github.com/MoinIhtejaz/Accenture_Datathon/blob/Raiha/docs/final_storyline.md	Pitch architecture	https://github.com/MoinIhtejaz/Accenture_Datathon/blob/Raiha/docs/final_strategy/final_pitch_storyboard.md
Claim status and boundaries	https://github.com/MoinIhtejaz/Accenture_Datathon/blob/Raiha/docs/final_strategy/evidence_register.md	Financial case	https://github.com/MoinIhtejaz/Accenture_Datathon/blob/Raiha/docs/final_strategy/financial_business_case.md
Implementation	https://github.com/MoinIhtejaz/Accenture_Datathon/blob/Raiha/docs/final_strategy/implementation_plan.md	Ethics and governance	https://github.com/MoinIhtejaz/Accenture_Datathon/blob/Raiha/docs/final_strategy/ethics_governance.md
Root-cause roles	https://github.com/MoinIhtejaz/Accenture_Datathon/blob/Raiha/docs/final_strategy/root_cause_classification.md	Technical appendix	https://github.com/MoinIhtejaz/Accenture_Datathon/blob/Raiha/docs/final_report/appendix/technical_evidence_appendix.md
HiPo concentration	https://github.com/MoinIhtejaz/Accenture_Datathon/blob/Raiha/docs/final_report/tables/table_01_hipo_concentration.csv	Priority strata	https://github.com/MoinIhtejaz/Accenture_Datathon/blob/Raiha/docs/final_report/tables/table_02_priority_strata.csv
Hypothesis verdicts	https://github.com/MoinIhtejaz/Accenture_Datathon/blob/Raiha/docs/final_report/tables/table_03_hypothesis_verdicts.csv	Agency economics	https://github.com/MoinIhtejaz/Accenture_Datathon/blob/Raiha/docs/final_report/tables/table_04_agency_economics.csv
Financial scenarios	https://github.com/MoinIhtejaz/Accenture_Datathon/blob/Raiha/docs/final_report/tables/table_05_financial_scenarios.csv	KPIs	https://github.com/MoinIhtejaz/Accenture_Datathon/blob/Raiha/docs/final_report/tables/table_07_kpis.csv
Value-pool sensitivity	https://github.com/MoinIhtejaz/Accenture_Datathon/blob/Raiha/outputs/tables/value_leakage_analysis.csv	Root-cause validation summary	https://github.com/MoinIhtejaz/Accenture_Datathon/blob/Raiha/outputs/tables/hipo_root_cause/summary.json
HiPo outcome concentration	https://github.com/MoinIhtejaz/Accenture_Datathon/blob/Raiha/outputs/tables/hipo_root_cause/hipo_outcome_concentration.csv		